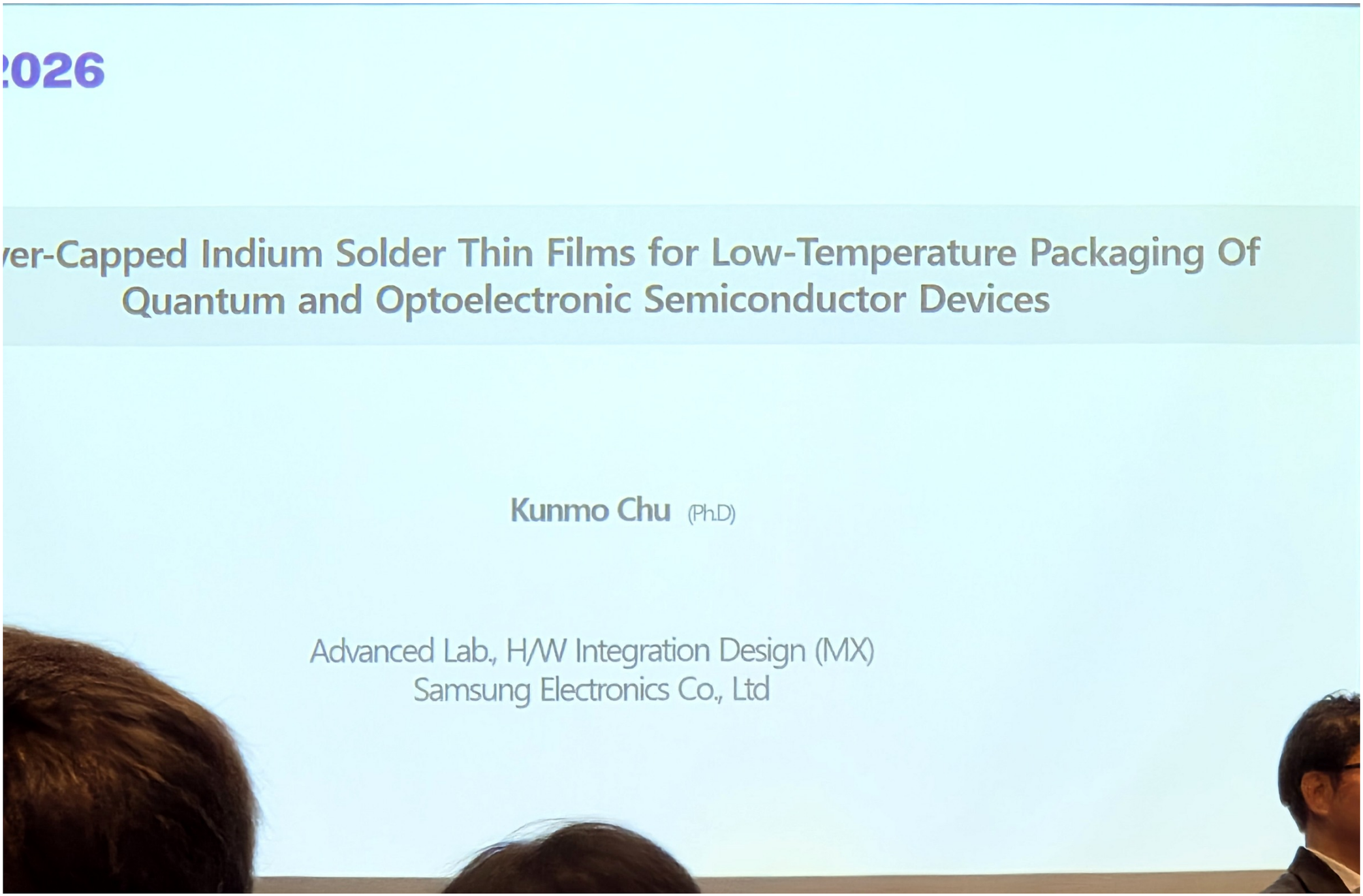
T37 은 캡핑 인듐의 저온·극저온 광전자 접합

Samsung Electronics · MX · 주건모 · Kunmo Chu

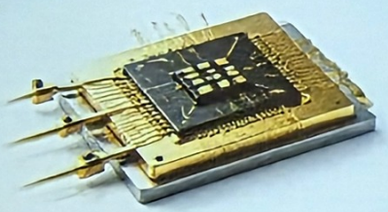


Background

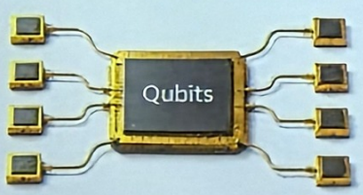
Quantum Processors Need Scalable Interconnects

As Quantum processors become larger, the number of connections increases dramatically.

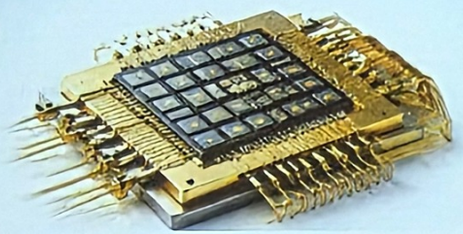
Early prototype
O(10) qubits



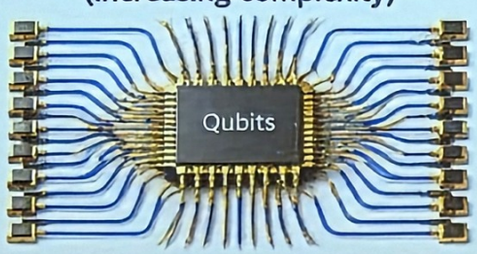
Few connections
(easy to manage)



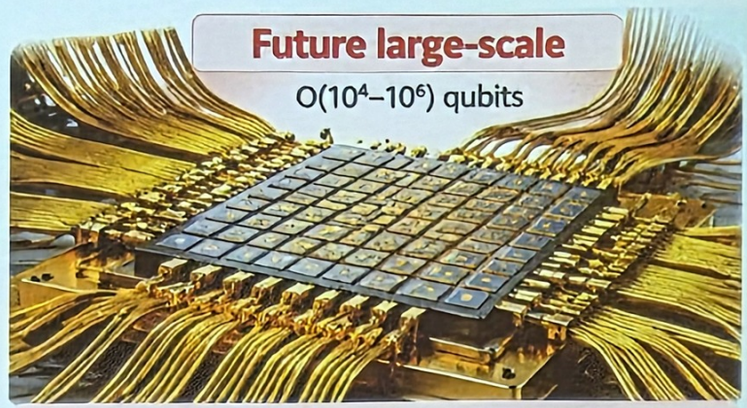
Intermediate-scale
O(10^2–10^3) qubits



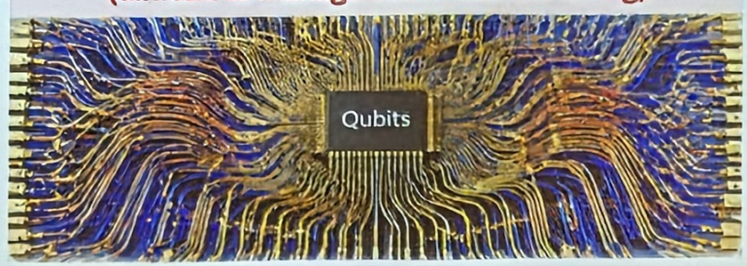
Hundreds–thousands of connections
(increasing complexity)



Future large-scale
O(10^4–10^6) qubits



Tens of thousands–millions of connections
(difficult to manage with electrical wiring)

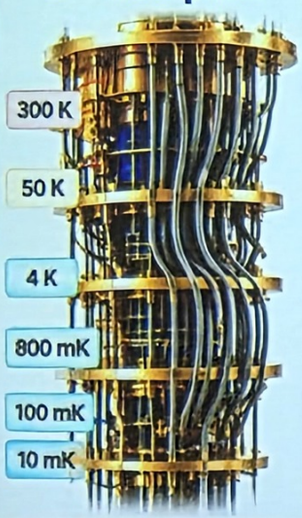


Background

Why Optical interconnect ?

As Quantum system scale, electrical interconnects become a major bottleneck.

01 Wiring Footprint



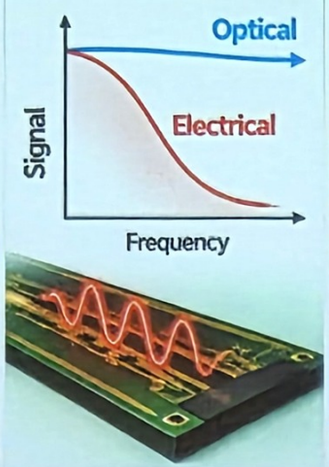
Space limit for scaling

02 Thermal Load



Unwanted heat to cold stage

03 Bandwidth



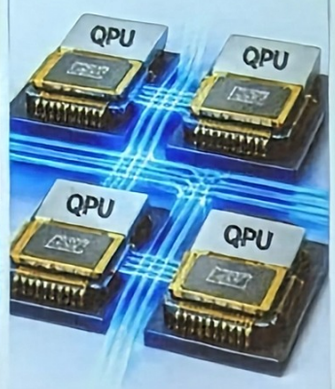
High-speed signal degradation

04 Crosstalk



Noise limits fidelity

05 Scalability



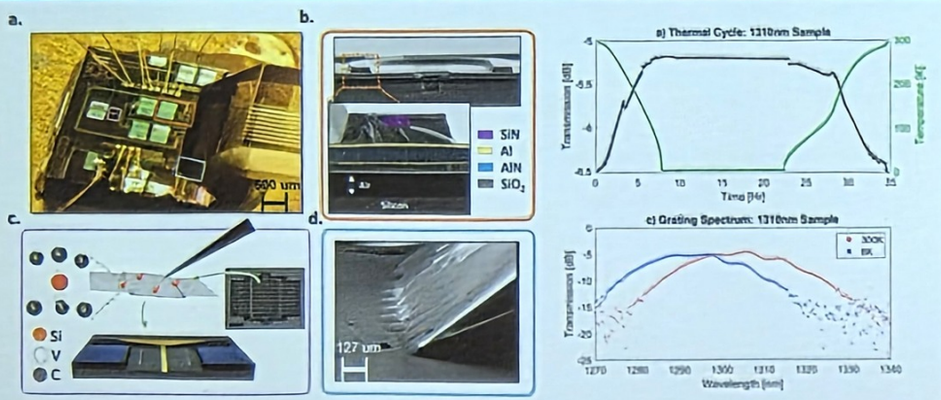
Need scalable chip-to-chip links

Bench mark

Scalable Cryogenic Optical Interfaces

Overcoming the Cryogenic Bottleneck: Low-Loss Photonic Packaging and Record-Fast Optical Data Links

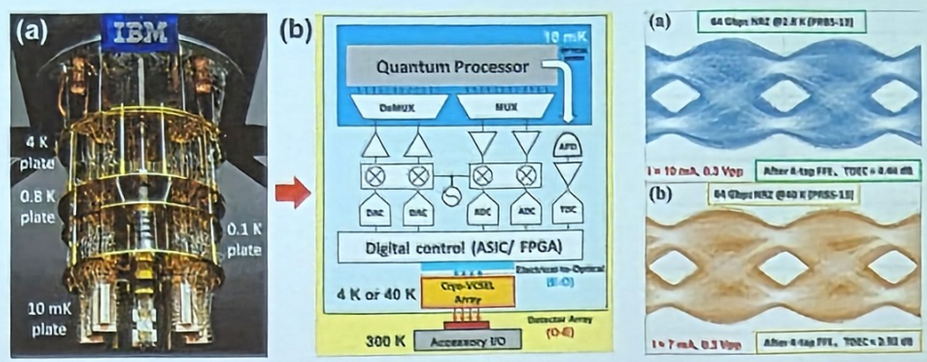
Tyndall National Institute



Target	• Quantum memory modules
Results	• Surface grating couplers & Angle-polished fiber arrays • Low loss: 2.2 dB per visible-wavelength grating coupler. • A reversible blueshift btw 300K and 6K.

¶ *Optica Quantum.* (2025) 3:569-575

UIUC



Target	• Cryo-VCSEL based optical data links
Results	• Delivers >13 mW output and >50 GHz bandwidth. • Demonstrates 112 Gb/s PAM-4 & 64 Gb/s NRZ links. • Reduces electrical wiring and thermal load.

¶ *IEEE J. Quantum Electron.* (2024) 60:2400310

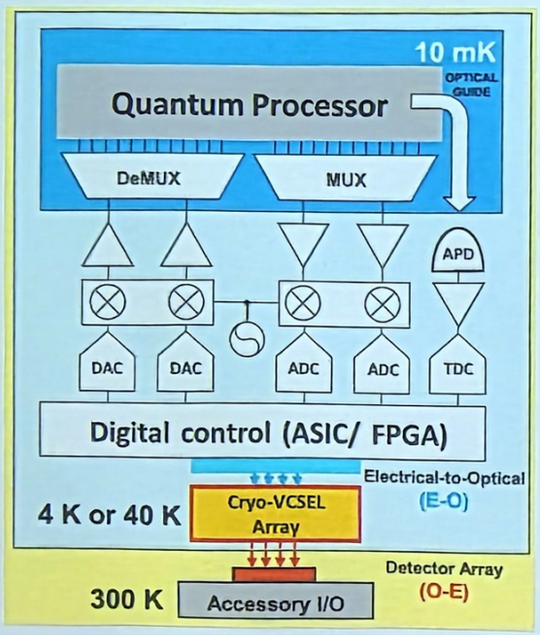
T37 은 캡핑 인듐의 저온·극저온 광전자 접합

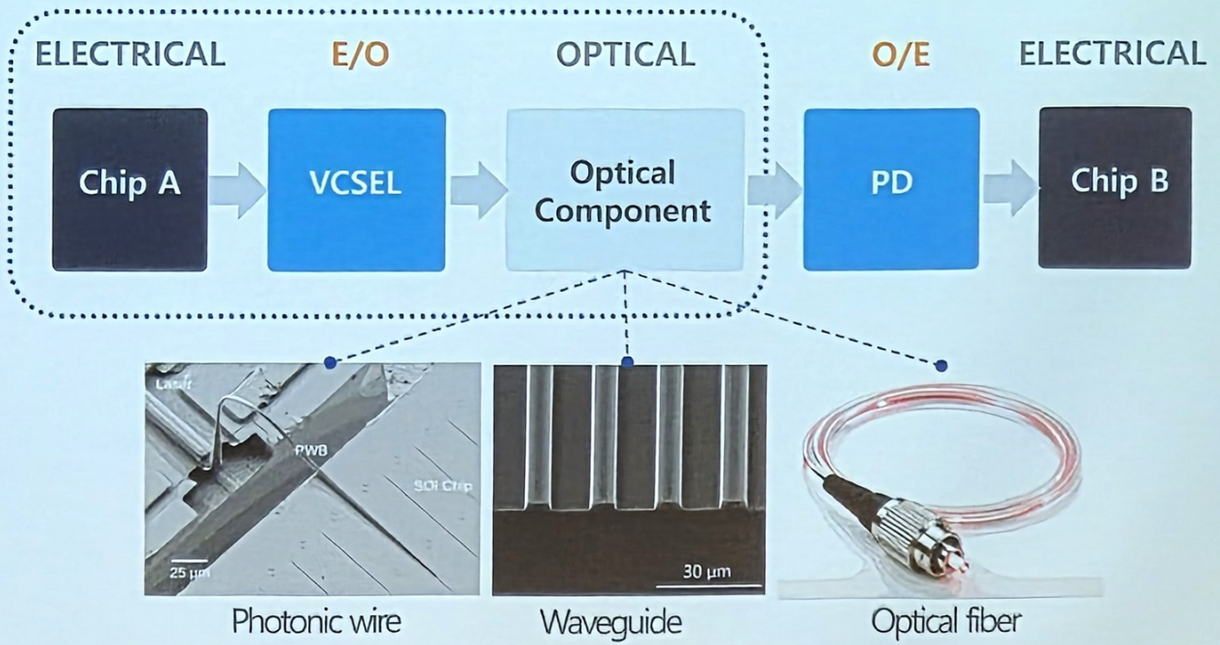
Samsung Electronics · MX · 주건모 · Kunmo Chu

Introduction

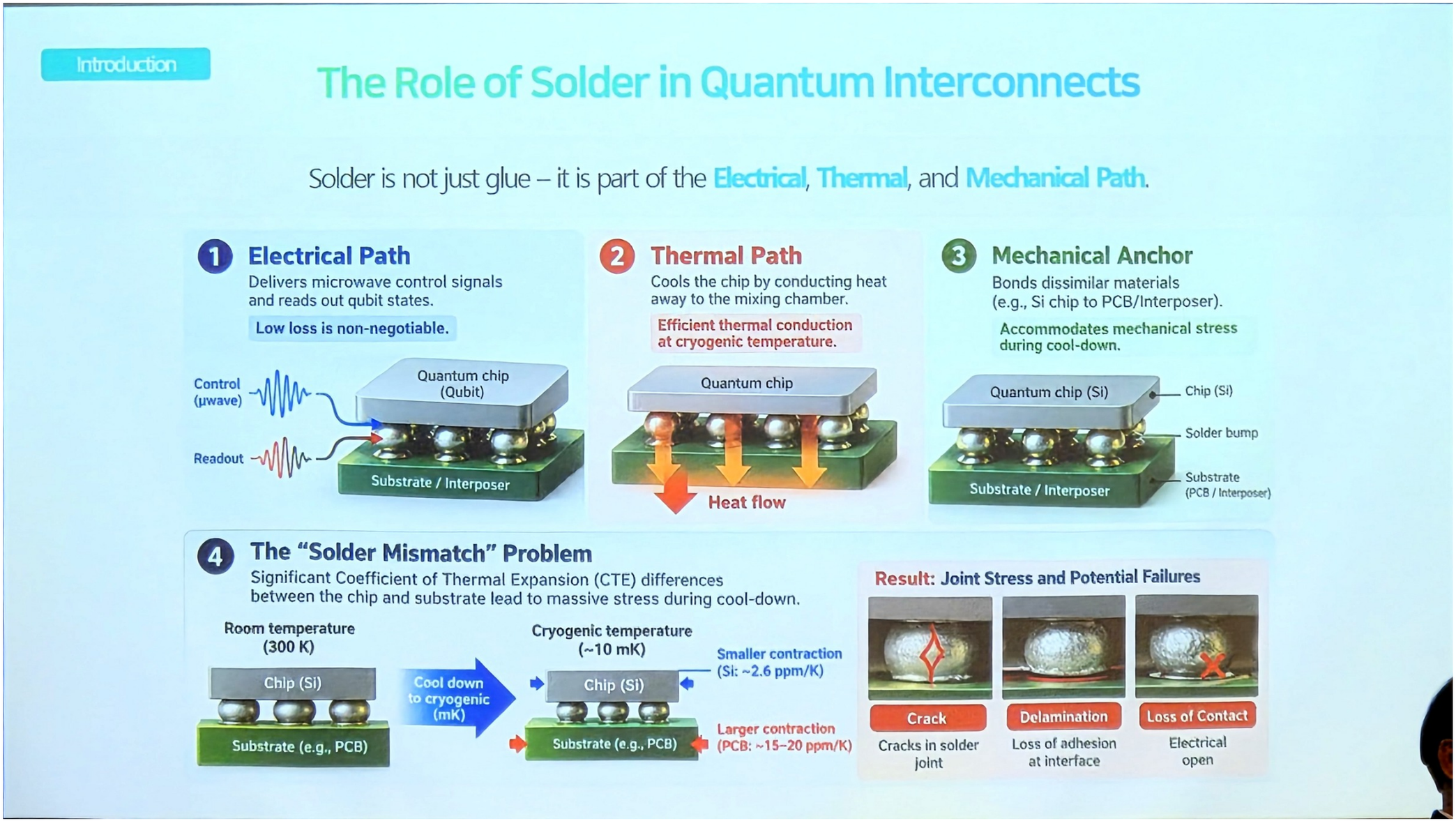
Target Architecture

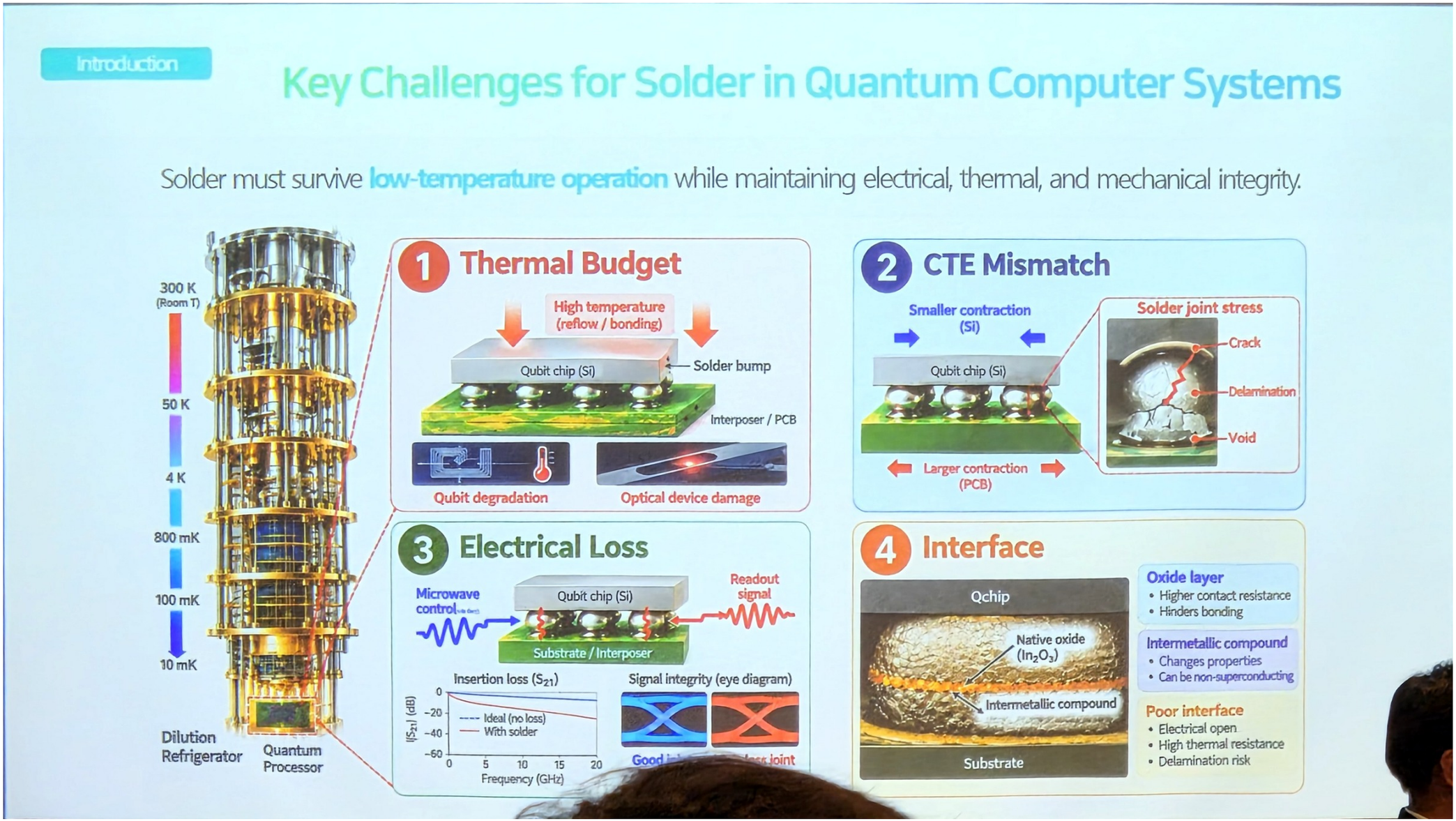
Cryogenic **VCSEL-based optical links** can replace heat-intensive, low-scalability coaxial wiring.





¶ *IEEE J. Quantum Electron.* 60 (2024) 60: 2400310


Introduction
The Role of Solder in Quantum Interconnects
Solder is not just glue – it is part of the Electrical, Thermal, and Mechanical Path.
1 Electrical Path
Delivers microwave control signals and reads out qubit states.
Low loss is non-negotiable.
Control (μwave)
Readout
Quantum chip (Qubit)
Substrate / Interposer
2 Thermal Path
Cools the chip by conducting heat away to the mixing chamber.
Efficient thermal conduction at cryogenic temperature.
Quantum chip
Heat flow
3 Mechanical Anchor
Bonds dissimilar materials (e.g., Si chip to PCB/Interposer).
Accommodates mechanical stress during cool-down.
Quantum chip (Si)
Chip (Si)
Solder bump
Substrate / Interposer
Substrate (PCB / Interposer)
4 The "Solder Mismatch" Problem
Significant Coefficient of Thermal Expansion (CTE) differences between the chip and substrate lead to massive stress during cool-down.
Room temperature (300 K)
Chip (Si)
Substrate (e.g., PCB)
Cool down to cryogenic (mK)
Cryogenic temperature (~10 mK)
Chip (Si)
Substrate (e.g., PCB)
Smaller contraction (Si: ~2.6 ppm/K)
Larger contraction (PCB: ~15–20 ppm/K)
Result: Joint Stress and Potential Failures
Crack
Cracks in solder joint
Delamination
Loss of adhesion at interface
Loss of Contact
Electrical open


Introduction
Key Challenges for Solder in Quantum Computer Systems
Solder must survive low-temperature operation while maintaining electrical, thermal, and mechanical integrity.
300 K (Room T)
50 K
4 K
800 mK
100 mK
10 mK
Dilution Refrigerator
Quantum Processor
1 Thermal Budget
High temperature (reflow / bonding)
Qubit chip (Si)
Solder bump
Interposer / PCB
Qubit degradation
Optical device damage
2 CTE Mismatch
Smaller contraction (Si)
Qubit chip (Si)
Larger contraction (PCB)
Solder joint stress
Crack
Delamination
Void
3 Electrical Loss
Microwave control
Qubit chip (Si)
Readout signal
Substrate / Interposer
Insertion loss (S_{21})
Ideal (no loss)
With solder
Frequency (GHz)
Signal integrity (eye diagram)
4 Interface
Qchip
Native oxide (In_2O_3)
Intermetallic compound
Substrate
Oxide layer
• Higher contact resistance
• Hinders bonding
Intermetallic compound
• Changes properties
• Can be non-superconducting
Poor interface
• Electrical open
• High thermal resistance
• Delamination risk

Introduction

Solder Candidates

Indium combines a low process temperature with high **Ductility** and intrinsic **Superconductivity** at **Cryogenic Temperature**.

Solder	Melting point	Low-temp process	Ductility / Compliance	Cryogenic relevance	Main concern
Sn58Bi	138 ℃	Strong	Brittle / limited ductility	Low-temp solder	Brittleness
Indium	**156.7 ℃**	**Strong**	**Excellent ductility**	**Superconducting below 3.4K**	**Oxide / Creep**
SAC305	217-220 ℃	Limited	Moderate	Common Pb-free solder	High thermal budget
AuSn	280 ℃	Limited	Hard / high strength	Can work at cryogenic T	Very high process T

Motivation

Why Indium ?

Indium is a strong candidate for interconnect material because of its **Low Melting Temp.** and **High Ductility.**

1 **LOW MELTING POINT**

156.7°C

Lower thermal budget than SAC305 and AuSn.

2 **HIGH DUCTILITY**

Soft + compliant

Accommodates CTE-driven strain during cooldown.



3 **CRYOGENIC BEHAVIOR**

Ductile at low T

Useful for wide temperature excursions.

4 **SUPERCONDUCTIVITY**

Tc ≈ 3.4 K

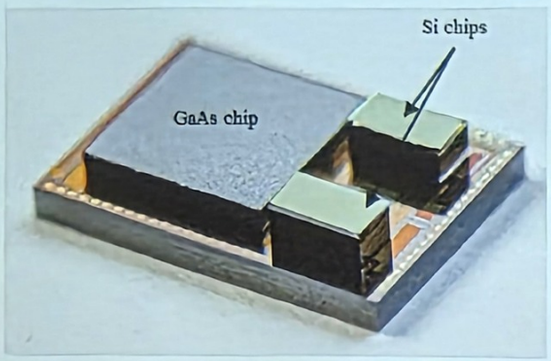
Useful when the interconnect is part of a superconducting path.

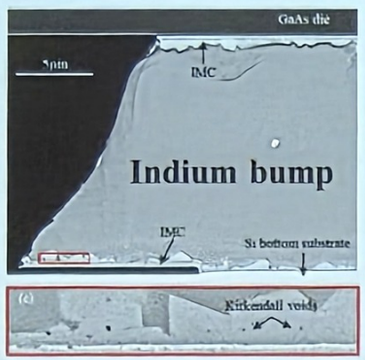
Bench mark

Indium Interconnects for Quantum Computing (1/2)

Indium is a promising material for low-temperature and quantum-related packaging

CEA-Leti

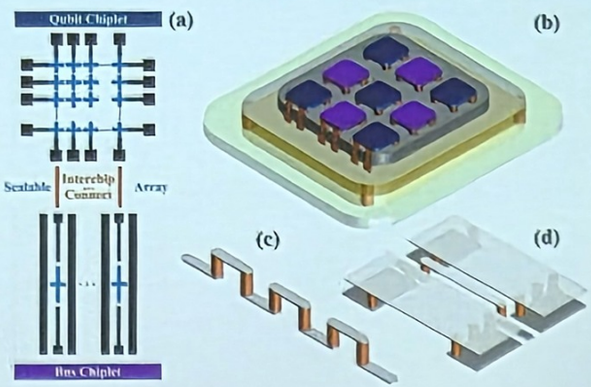


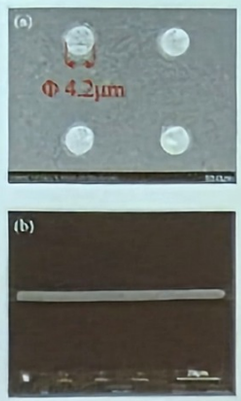


Target	• Multi-chip Quantum Platform (QuIC³)
Results	• Indium microbumps for 3D Si-Si and GaAs packaging. • Stable electrical and mechanical connections • Au-In IMCs and Kirkendall voids were observed

¶ *2023 IEEE 73rd ECTC*, pp. 973-979

Tsinghua Univ.





Target	• Quantum 3D Integration
Results	• Reliable In-In bonding above 40 MPa. • 4.2 μm bumps with >15.2 MPa shear strength. • Stable performance from 300 K to 2 K.

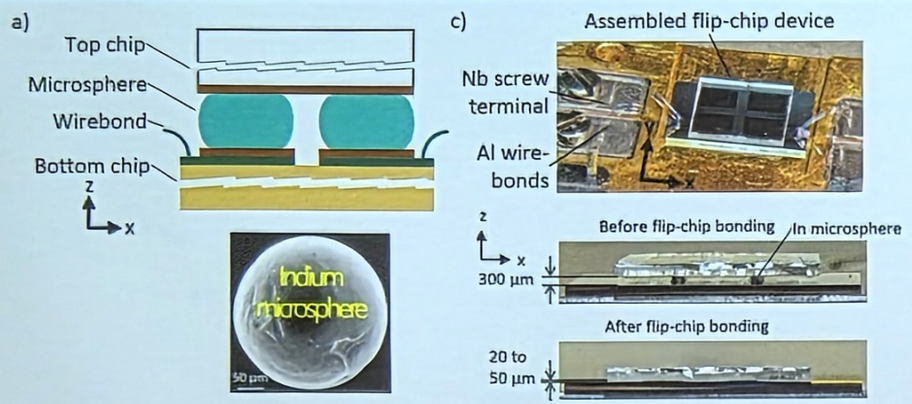
¶ *IEEE Electron Device Lett.* (2024) 45:2021--2024

Bench mark

Indium Interconnects for Quantum Computing (2/2)

Indium provides both electrical continuity and mechanical compliance at cryogenic temperature..

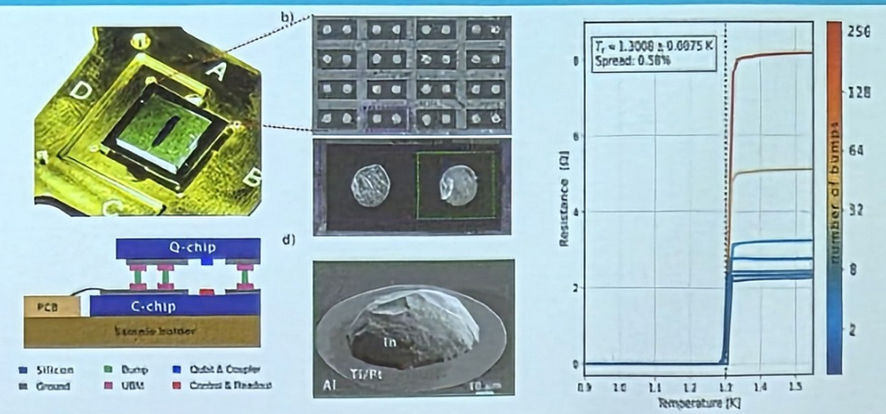
Chalmers Univ. of Tech.



Target	• 3D Quantum and Cryogenic devices
Results	• Flip-chip bonding with indium microspheres (300 µm) • UBM: Au-passivated Nb or NbN • Simple & high-current interconnects possible

¶ *Appl. Phys. Lett.* (2025) 126:022601

PSL Research Univ.



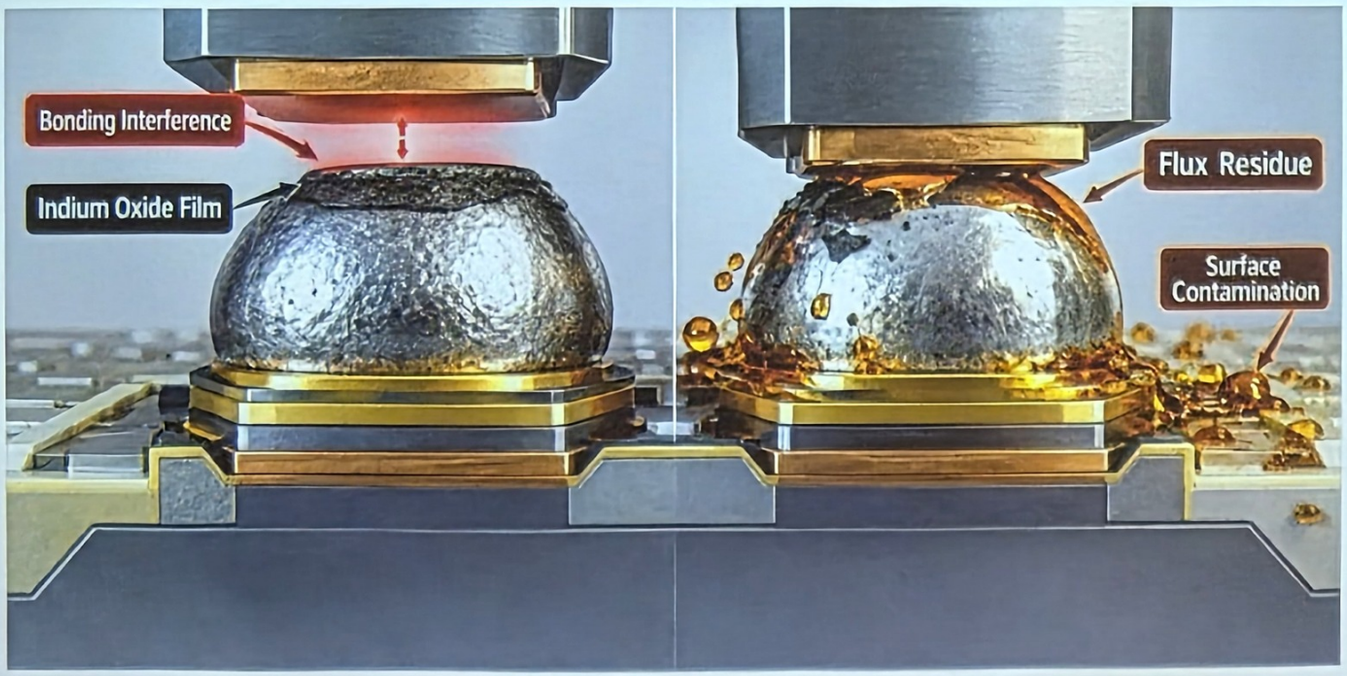
Target	• Superconducting Quantum Processors
Results	• Indium reduce thermal stress between Q and C-chips. • UBM: 5 nm Ti / 5 nm Pt • Al/In interface can affect the co

¶ *Adv. Quantum Technol.* (2025) 8:e2500212

Motivation

But, Indium Is Not Perfect

Oxide, wetting, creep, and intermetallic formation must be managed.



Oxide
Can block intimate metal contact

Flux
Can contaminate the components

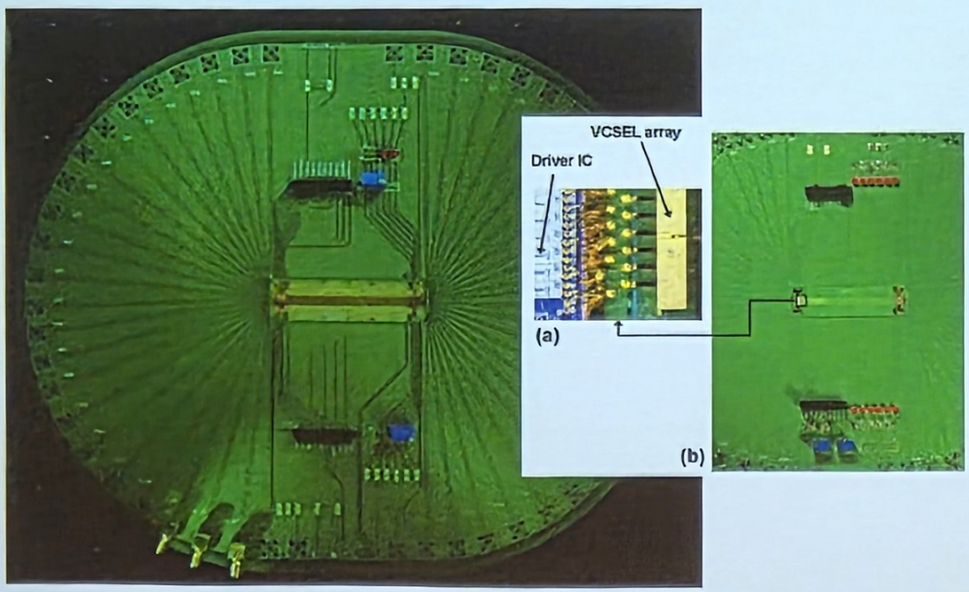
Creep
Softness helps compliance but can affect long-term shape

Suggestion

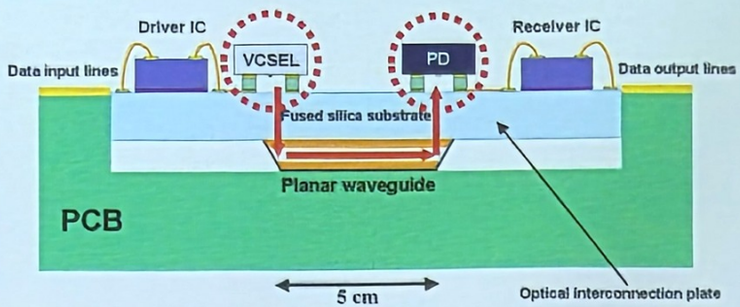
Proposed Package Structure

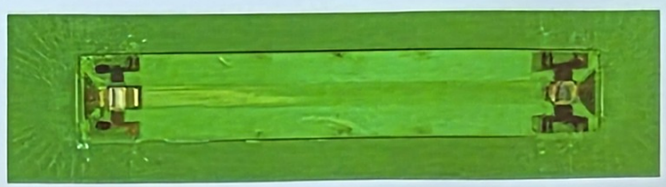
The optical interconnection plate is placed between input and output data lines on PCB that requires an optical link.

PCB for Chip-to-Chip Optical Interconnection



Optical Interconnection Plate



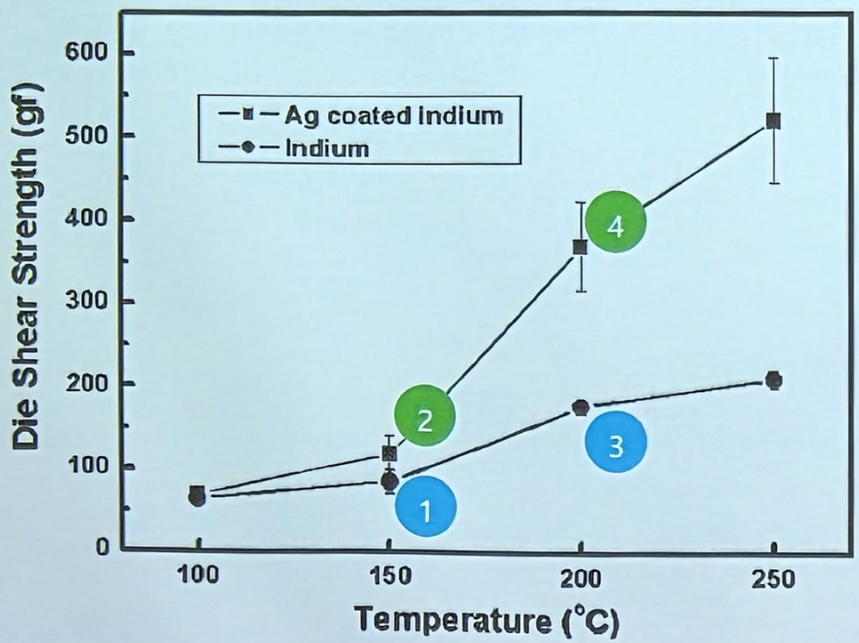


Result

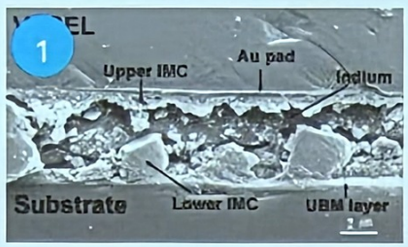
Silver(Ag) coating effects (3/4): Bonding Strength

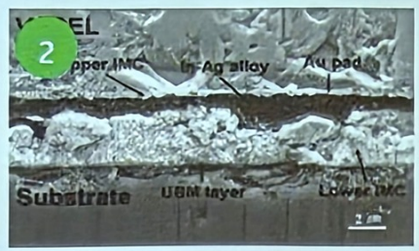
Thin Ag layer increases the **Die shear strength**, while the joint microstructure remains similar.

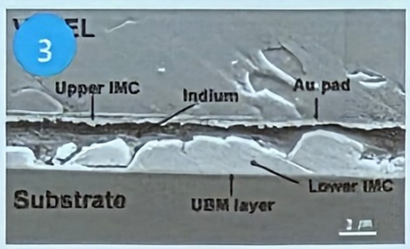
The Die Shear Test

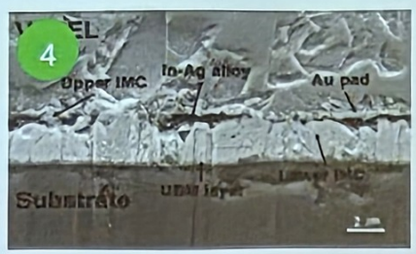


Joint Microstructure Analysis







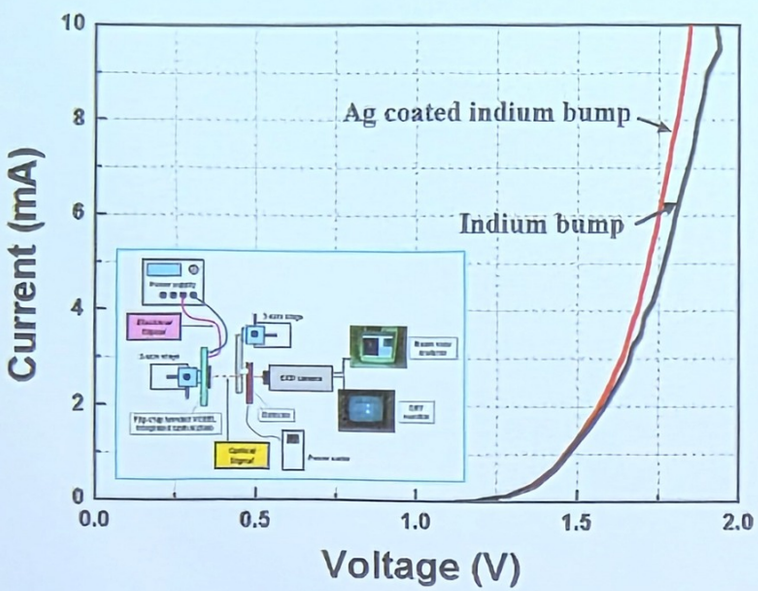


Result

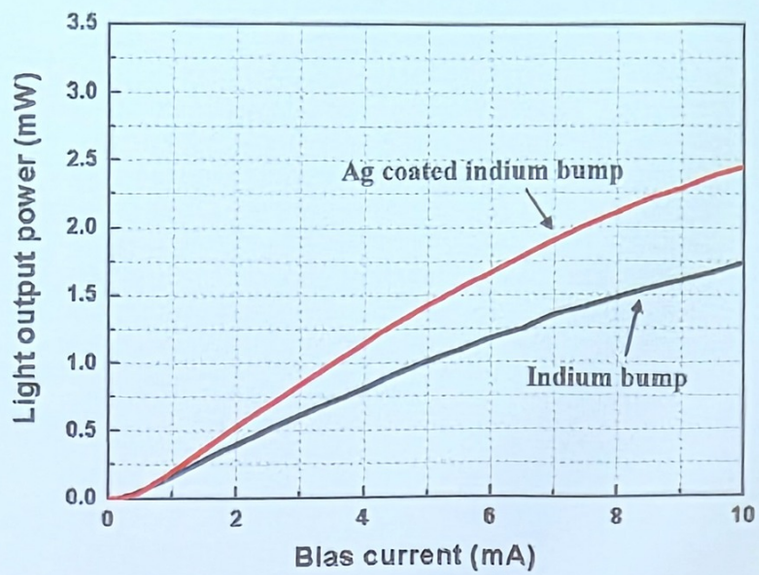
Silver(Ag) coating effects (4/4): Opto-Electrical Characteristics

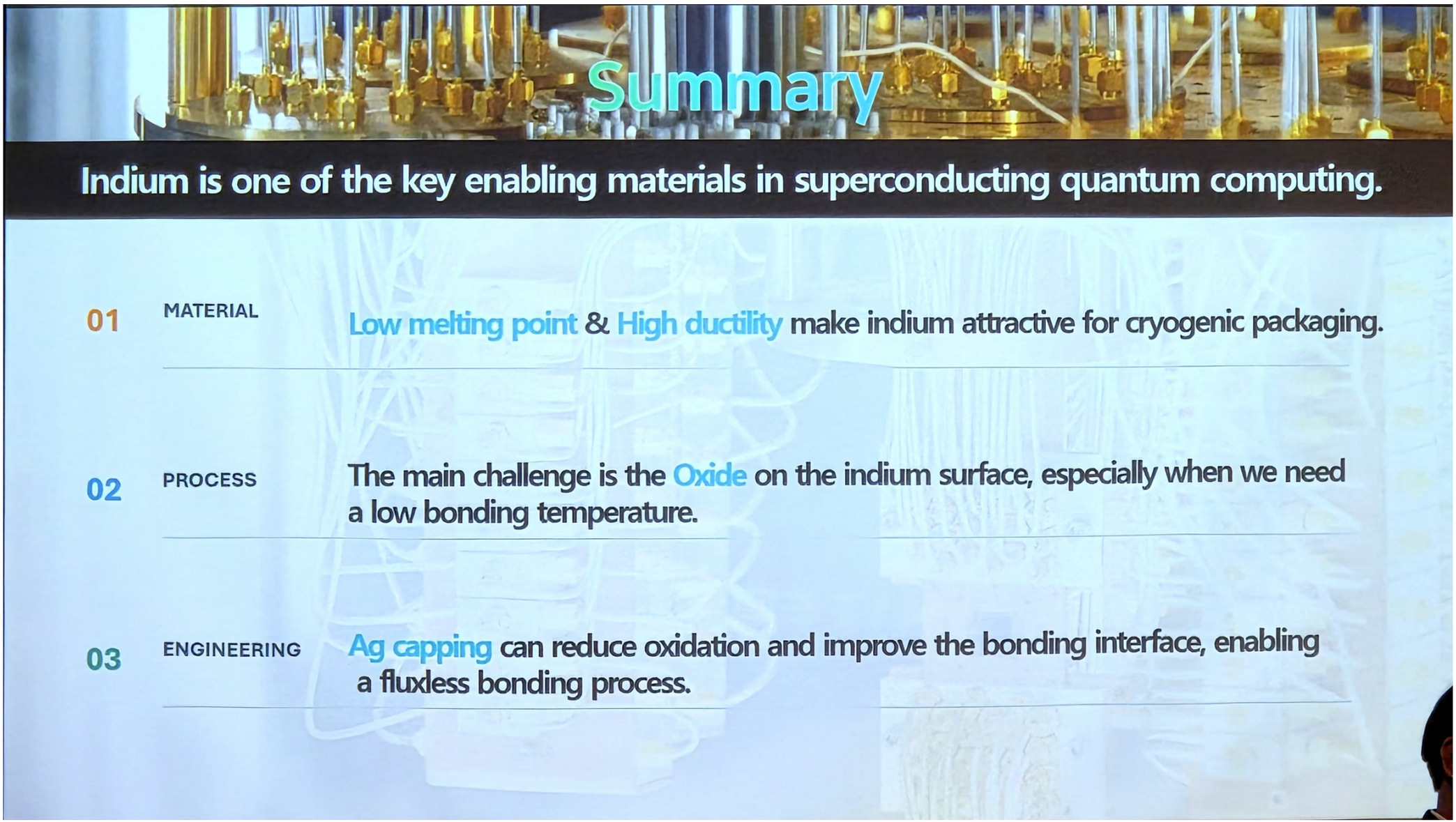
Thin Ag layer improves the **Electrical Contact Interface** and reduces **Series Resistance**.

I-V curves



L-I curves




Summary
Indium is one of the key enabling materials in superconducting quantum computing.
01
MATERIAL
Low melting point & High ductility make indium attractive for cryogenic packaging.
02
PROCESS
The main challenge is the Oxide on the indium surface, especially when we need a low bonding temperature.
03
ENGINEERING
Ag capping can reduce oxidation and improve the bonding interface, enabling a fluxless bonding process.